Question 1: Manually trace the following code and show changes to variables:

```
int count = 0;
for (int i = 1; i <= 5; i++) {
    for (int j = i; j <= 5; j++) {
        count += j;
        printf (“\n I = %d\t J = %d\t Count = %d”, i, j, count);
    }
}
printf("\n\nCount: %d", count);
```

Question 2: Manually trace the following code and show changes to variables:

```
int total = 0;
for (int m = 1; i <= 3; i++) {
    for (int n = 1; j <= 4; j++) {
        total += i * j - 1;
        printf(“\n %d\t %d\t %d”, m, n, total);
    }
}
printf("Total: %d", total);
```

Question 3

Manually trace the given code segment. Show the changes of all the variables i and array elements arr in each step.

```
int arr[5] = {1, 2, 3, 4, 5};
for(int i = 0; i < 5; i++) {
   arr[i] = arr[i] * 2;

  printf(“\n i = %d\t arr[%d] =%d”, i, i, arr[i]);
}
for(int i = 4; i >= 0; i--) {
   arr[i] = arr[i] - 1;

 printf(“\nArrary[%d] = %d”, i, arr[i]);
}
```